

Week 9

4.1 Orthonormality

def 4.10 A list of vectors is called **orthonormal** if every vector in the list has norm 1 and is orthogonal to every other vector in the list

lemma 4.12 If $e_1, \dots, e_n$ is a list of orthonormal vectors in an inner product space V , then the norm

$$\|a_1 e_1 + \dots + a_n e_n\|^2 = |a_1|^2 + \dots + |a_n|^2$$

for any $a_i \in \mathbb{F}$

def 4.13 An **orthonormal basis** is a basis for an inner product space V that is orthonormal

cor. 4.14 Let $\{e_i \mid i \in I\}$ be an orthonormal list of vectors in an inner product space V . Then the list is linearly independent

cor 4.15 For a finite dimensional inner product space V , if $\{e_1, \dots, e_n\}$ is a list of basis vectors of length $\dim V$, then it is a basis

lemma 4.16 if V is an inner product space of finite dimension n , and $\{e_1, \dots, e_n\}$ is an orthonormal basis for V , then for any $v \in V$,

$$v = \langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2 + \dots + \langle v, e_n \rangle e_n$$

thm 4.17 **Gram-Schmidt** - Let $v_1, \dots, v_m$ be a linearly independent list of vectors in V . Then, let $e_1 = \frac{v_1}{\|v_1\|}$ and inductively define

$$e_{j+1} = \frac{v_j - \langle v_j, e_1 \rangle e_1 - \dots - \langle v_j, e_j \rangle e_j}{\|v_j - \langle v_j, e_1 \rangle e_1 - \dots - \langle v_j, e_j \rangle e_j\|}$$

then $e_1, \dots, e_m$ is an orthonormal list of vectors in V & $\text{span}_{\mathbb{F}}(v_1, \dots, v_j) = \text{span}_{\mathbb{F}}(e_1, \dots, e_j)$ for each j

lemma 4.18 If $e_1, \dots, e_m$ is an orthonormal list in a finite-dimensional vector space V of dimension n , then there are vectors $f_{m+1}, \dots, f_n$ such that $e_1, \dots, e_m, f_{m+1}, \dots, f_n$ is an orthonormal basis for V .

lemma 4.19 Let $\alpha: V \rightarrow V$ be a linear map of finite dimensional complex inner product spaces. Then, if there is a basis of V with respect to which α is upper triangular, we may take that basis to be orthonormal. In particular, if V is complex as a vector space, then there is an orthonormal basis with respect to which α is upper-triangular.

thm 4.20 **Riesz Representation Theorem** - Let V be a finite dimensional inner product space and let $\varphi: V \rightarrow \mathbb{F}$ be a linear map (so $\varphi \in V^*$). Then, there is a unique vector $u \in V$ st for every $v \in V$

$$\varphi(v) = \langle v, u \rangle$$

Orthogonal Components

def 4.21 In an inner product space $(V, \langle \cdot, \cdot \rangle)$, if U is a subset of V , the orthogonal complement of U is given by $U^\perp = \{v \in V, \langle v, u \rangle = 0 \text{ for all } u \in U\}$

lemma 4.22 If U is a subset of an inner product space V , then:

(a) if U is a subset of V , then $U^\perp$ is a subspace of V

(b) $\{0\}^\perp = V$

(c) $V^\perp = \{0\}$

(d) if U is a subset of V , then $U \cap U^\perp = \{0\}$

(e) if U and W are subsets of V and $U \subset W$, then $W^\perp \subset U^\perp$

lemma 4.23 Suppose U is a finite-dimensional subspace of V . Then, $V = U \oplus U^\perp$